very well (and it would translate into a linear search for less 1001 elements!).

In practice, through profiling and performance tests, it has been reached the conclusion that in most situations, `D=4` is the best compromise.

2.9.3 Finding the optimal branching factor

If you are looking for an optimal value for `D` that works in every situation, then you are going to be disappointed. To a certain extent, theory comes to the rescue showing us a range of optimal values. It can be shown that the optimal value can't be greater than 5. Or, to put it another way, it can be mathematically proven that:

- The tradeoff between insert and delete is best balanced with `2 <= D <= 5`;
- A 3-way heap is in theory faster than 2-way heap;
- 4-way heaps and 3-way heaps have similar performance;
- 5-way heaps are a bit slower.

In practice, the best value for `D` depends on the details of the implementation and of the data you are going to store in the heap. The optimal branching factor for a heap can only be determined empirically, case-by-case: there is no overall optimal branching factor, and it depends on the actual data and on the ratio of insertions/deletions, or f.i., how expensive it is to compute priority versus copying elements, among other things.